

# CHAPTER 12

# TRANSFER OF

# THERMAL ENERGY

## 2.3 Transfer of thermal energy

### 2.3.1 Conduction

- 1 Describe experiments to distinguish between good and bad thermal conductors
- 2 Describe thermal conduction in all solids in terms of atomic or molecular lattice vibrations and also in terms of the movement of free (delocalised) electrons in metallic conductors

### 2.3.2 Convection

- 1 Explain convection in liquids and gases in terms of density changes and describe experiments to illustrate convection

### 2.3.3 Radiation

- 1 Describe the process of thermal energy transfer by infrared radiation and know that it does not require a medium
- 2 Describe the effect of surface colour (black or white) and texture (dull or shiny) on the emission, absorption and reflection of infrared radiation
- 3 Describe how the rate of emission of radiation depends on the surface temperature and surface area of an object
- 4 Describe experiments to distinguish between good and bad emitters of infrared radiation
- 5 Describe experiments to distinguish between good and bad absorbers of infrared radiation

### 2.3.4 Consequences of thermal energy transfer

- 1 Explain everyday applications using ideas about conduction, convection and radiation, including:
  - (a) heating objects such as kitchen pans
  - (b) heating a room by convection
  - (c) measuring temperature using an infrared thermometer
  - (d) using thermal insulation to maintain the temperature of a liquid and to reduce thermal energy transfer in buildings

**1. O/N/23/11/Q16**

A student on a camping expedition cools a sealed bottle of water which is at the same temperature as the surrounding air.

Which method cools the water at the greatest rate?

- A** Wrap the bottle in aluminium foil and place it in a shady place.
- B** Wrap the bottle in dry, white paper and put it in a sunny place.
- C** Wrap the bottle in foam and put it in a breeze.
- D** Wrap the bottle in wet paper and put it in a breeze.

**2. O/N/23/11/Q17**

A student suggests three factors that affect the rate of emission of thermal energy by radiation from a hot object. These are:

- 1 the surface temperature of the object
- 2 the surface area of the object
- 3 the surface colour of the object.

Which suggestions are correct?

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

**3. O/N/23/12/Q21**

Solid metals are good thermal conductors.

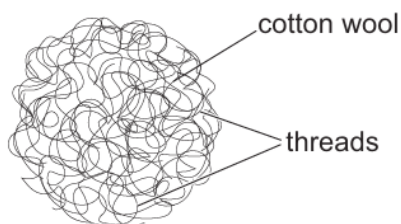
How closely packed are their particles and do the metals contain free electrons?

|          | packing of particles | contains<br>free electrons |
|----------|----------------------|----------------------------|
| <b>A</b> | close together       | no                         |
| <b>B</b> | close together       | yes                        |
| <b>C</b> | far apart            | no                         |
| <b>D</b> | far apart            | yes                        |

#### 4. O/N/23/12/Q22

Cotton wool is a good thermal insulator.

It consists of many threads of cotton which are tangled together to make a material which can easily be pulled apart or compressed. The diagram shows the structure of cotton wool.

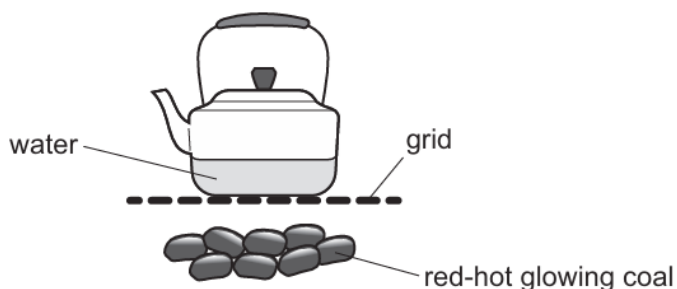


Why is cotton wool a good thermal insulator?

- A Each thread is very thin.
- B It has a large surface area because it is made of many threads.
- C It has a low density because of the trapped air.
- D It traps air and prevents convection.

#### 5. O/N/23/12/Q23

A metal kettle is heated on a grid above red-hot glowing coal.

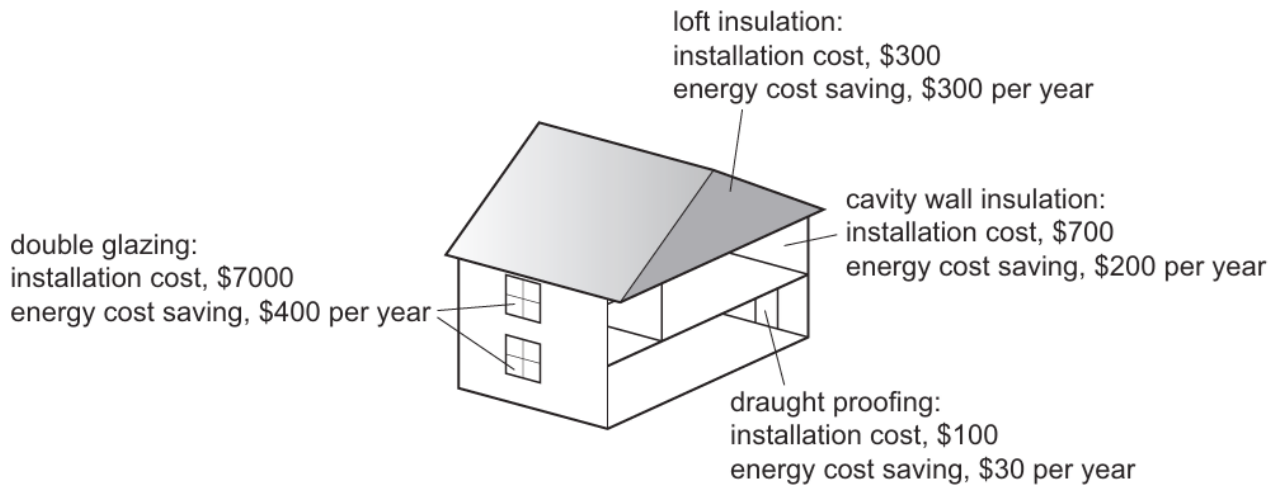


What are the main thermal energy transfers through the bottom of the kettle and within the water?

|   | through kettle | within water |
|---|----------------|--------------|
| A | conduction     | conduction   |
| B | radiation      | convection   |
| C | conduction     | convection   |
| D | radiation      | conduction   |

**6. M/J/23/11/Q18**

A house is to be thermally insulated to reduce thermal energy loss and reduce fuel costs.



Which type of thermal insulation pays back the cost of installation in the shortest time?

- A** cavity wall insulation
- B** double glazing
- C** draught proofing
- D** loft insulation

**7. M/J/23/12/Q17**

Which statement about copper explains why it is a better thermal conductor than glass?

- A** Atomic vibration is passed on to neighbouring copper atoms slowly.
- B** Atoms move through the copper and pass on kinetic energy.
- C** There are density changes within the copper.
- D** There are free electrons in the copper.

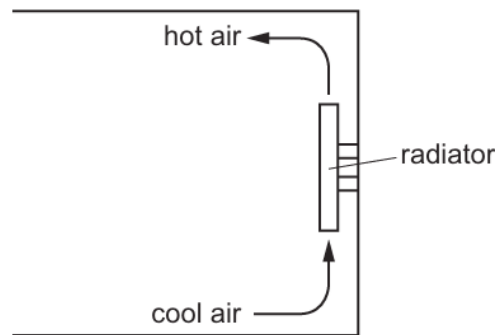
**8. M/J/23/12/Q18**

Which statement describes the transfer of thermal energy in a liquid by convection?

- A** A decrease in its density causes the heated liquid to rise.
- B** Free electrons carry energy large distances through the liquid.
- C** Infrared radiation passes through the liquid and transfers energy.
- D** Particles of the liquid vibrate and pass energy to neighbouring particles.

**9. O/N/22/11/Q17**

The diagram shows a radiator heating the air in a room.



What is the name of this process?

- A conduction
- B convection
- C evaporation
- D expansion

**10. O/N/22/11/O18**

What is a process of heat transfer that can take place in a vacuum?

- A conduction
- B convection
- C evaporation
- D radiation

**11. M/J/22/11/Q13**

A building has walls made from wood and steel.

What describes the way in which thermal energy is transferred through the walls?

- A conduction through the wood by movement of molecules through the wood only
- B conduction through the metal by both the vibration of particles and movement of electrons through the steel
- C conduction through the wood by both the vibration and movement of molecules through the wood
- D convection through the metal by movement of electrons through the steel only

**12. M/J/22/12/Q16**

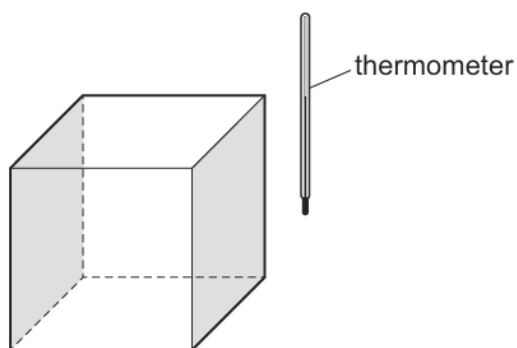
How is heat energy transferred through the vacuum of space from the Sun to the Moon's surface?

- A conduction only
- B convection only
- C radiation only
- D conduction, convection and radiation

**13. M/J/22/12/Q17**

A metal cube contains boiling water.

Each vertical face of the cube is painted a different colour or has a different texture.



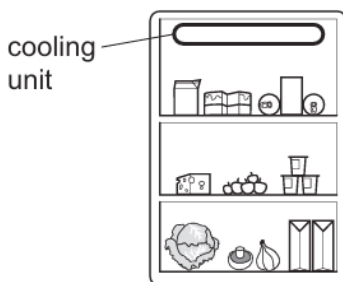
Identical thermometers are held at equal distances from each vertical face.

Near which face does a thermometer read the greatest temperature?

- A dull black
- B dull white
- C shiny black
- D shiny white

**14. O/N/21/11/Q12**

The diagram shows the inside of a refrigerator.



When the refrigerator is first switched on, what happens to the air near the cooling unit?

|   | the molecules of this air | the density of this air |
|---|---------------------------|-------------------------|
| A | become smaller            | decreases               |
| B | become smaller            | increases               |
| C | move closer together      | decreases               |
| D | move closer together      | increases               |

**15. O/N/21/12/Q15**

What is the colour and what is the texture of a good absorber of infrared radiation?

- A black and shiny
- B black and dull
- C white and shiny
- D white and dull

**16. M/J/21/11/Q15**

A copper rod is heated at one end.

Which statement describes how heat transfer occurs in the copper?

- A Energetic copper molecules move from the cooler end to the hotter end.
- B Energetic copper molecules move from the hotter end to the cooler end.
- C Energetic free electrons move from the cooler end to the hotter end.
- D Energetic free electrons move from the hotter end to the cooler end.

**17. M/J/21/12/Q18**

Which heat transfer processes do **not** require a medium?

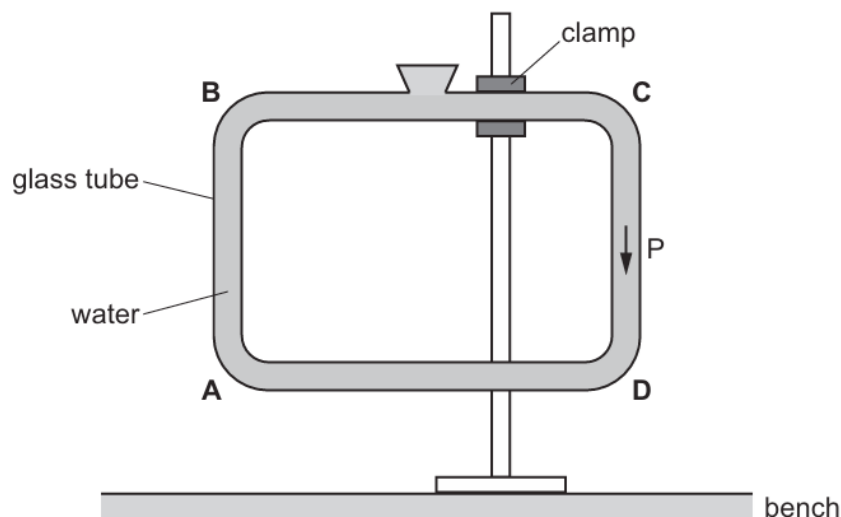
- A conduction only
- B convection only
- C radiation only
- D conduction and radiation

**18. O/N/20/12/Q17**

The diagram shows a glass tube filled with water and suspended above a bench. The water is free to circulate around the tube.

At point P, there is a convection current moving in a downwards direction.

At which point is the tube heated to cause this convection current?



**19. M/J/20/11/Q21**

Which description of a dull black surface is correct?

- A** good emitter, good absorber and good reflector of radiation
- B** good emitter, poor absorber and poor reflector of radiation
- C** good emitter, good absorber and poor reflector of radiation
- D** poor emitter, poor absorber and poor reflector of radiation

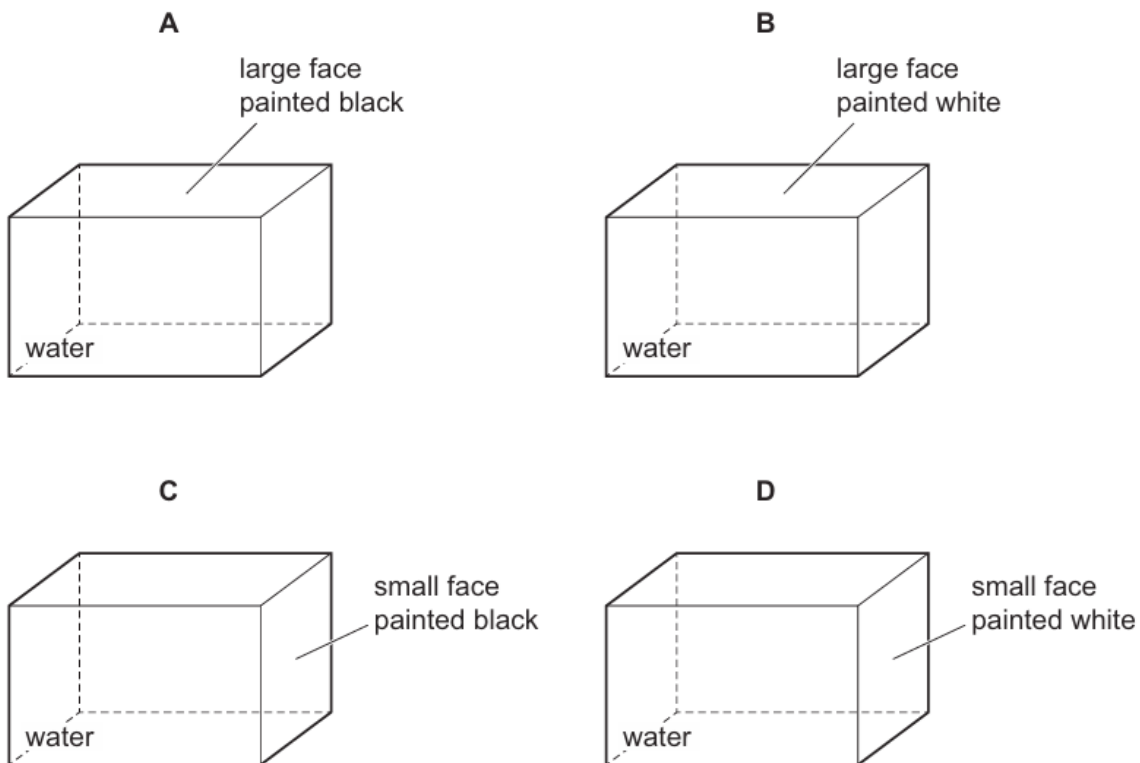
**20. M/J/20/12/Q23**

Four metal containers, with identical dimensions, are filled with water at 90 °C.

All the faces, except one, of each container are covered in a very good insulator.

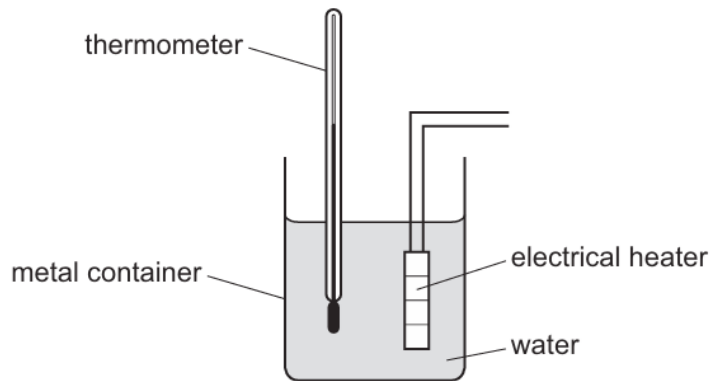
The one exposed face on each container is painted either black or white.

In which container does the water cool the fastest?



**21. M/J/19/11/Q17**

The diagram shows a set of apparatus used to determine the specific heat capacity of water.



What does **not** affect the rate at which energy is lost to the surroundings?

- A insulating the container
- B placing a lid on the container
- C polishing the outer surface of the container
- D moving the thermometer closer to the heater

**22. O/N/18/11/Q17**

A cold solid is placed on top of a hot solid. Thermal energy flows from the hot solid to the cold one.

What is the explanation for this?

- A A hot solid expands, so its molecules will move further apart.
- B Energy is passed from one molecule to the next.
- C Heat always rises.
- D Molecules are free to move randomly through the solids.

**23. O/N/18/12/Q17**

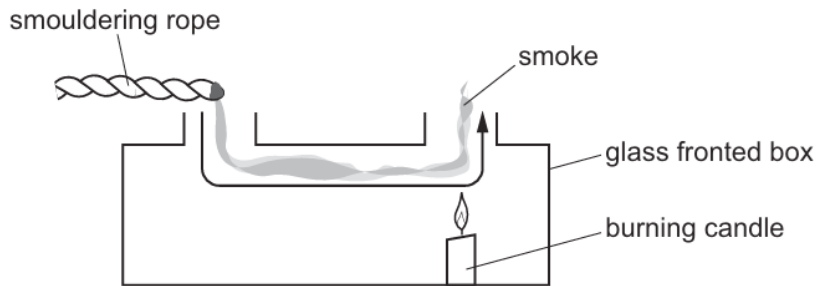
A cold bottle containing a drink is placed on a table on a warm day. Drops of water form on the outside of the bottle.

Which process causes the drops to form?

- A condensation
- B conduction
- C convection
- D evaporation

**24. M/J/18/11/Q20**

When a piece of smouldering rope is held at the opening of the box in the diagram, smoke moves in the direction indicated.



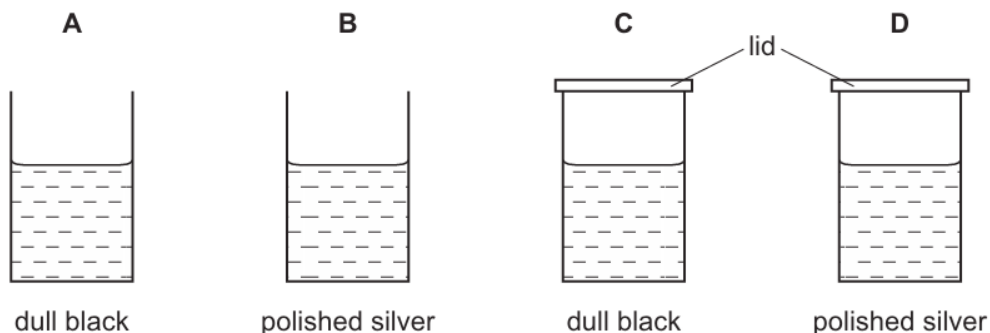
What is responsible for the movement of the smoke?

- A** convection
- B** movement of free electrons
- C** radiation
- D** vibration of molecules

**25. M/J/18/11/Q21**

The diagrams show four identical cans with their outside surfaces either polished silver or painted dull black. Each can contains the same volume of water, initially at 80 °C.

After five minutes in a cool room, which can contains the coolest water?



**26. M/J/18/12/Q24**

Samples of four different materials at room temperature are heated from below and heat is transferred upwards.

In which material is all of the heat transferred by the vibration of molecules?

- A** air
- B** mercury
- C** rubber
- D** water

**27. O/N/17/11/Q16**

In which substance is the conduction of thermal energy mainly due to the movement of electrons?

- A** air
- B** ice
- C** iron
- D** water

**28. O/N/17/11/Q17**

Four similar metal plates are the same distance from a heater that emits infra-red radiation.

The plates are painted dull black, dull white, shiny black and shiny white.

Which plate absorbs the most radiation and which plate reflects the most radiation?

|          | absorbs<br>most radiation | reflects<br>most radiation |
|----------|---------------------------|----------------------------|
| <b>A</b> | dull black                | dull white                 |
| <b>B</b> | dull black                | shiny white                |
| <b>C</b> | shiny black               | dull black                 |
| <b>D</b> | shiny white               | dull black                 |

**29. O/N/17/12/Q18**

A heater is designed to radiate thermal energy.

Which change to the design decreases the thermal energy emitted by radiation?

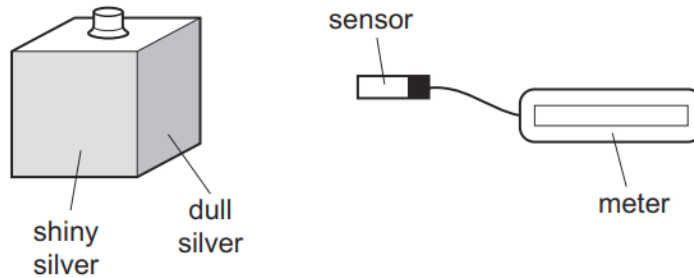
- A** a darker coloured surface
- B** a higher surface temperature
- C** a larger surface area
- D** a shinier surface

**30. M/J/17/12/Q25**

A metal box has four different surfaces; dull black, shiny black, dull silver and shiny silver.

The box is filled with boiling water so that each surface is at the same temperature.

A sensor measures the amount of radiation from each surface.



Which surface emits the least radiation and which surface emits the most radiation?

|          | least        | most         |
|----------|--------------|--------------|
| <b>A</b> | dull black   | shiny silver |
| <b>B</b> | dull silver  | shiny black  |
| <b>C</b> | shiny black  | dull silver  |
| <b>D</b> | shiny silver | dull black   |

**31. O/N/16/11/Q20**

Which statement about the transfer of thermal energy is correct?

- A** Transfer by radiation does not require a medium.
- B** Transfer is from a region of lower temperature to one of higher temperature.
- C** Transfer upwards in fluids is mainly through the vibrations of neighbouring particles.
- D** Transfer in solids is by means of density changes in the material.

**32. O/N/16/12/Q24**

In which states of matter does convection occur?

- A** gases, liquids and solids
- B** gases and liquids only
- C** gases and solids only
- D** liquids and solids only

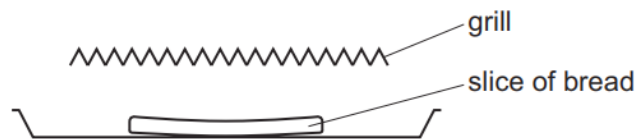
**33. M/J/16/11/Q18**

Which statement about thermal radiation is correct?

- A** In a vacuum, thermal radiation travels at the speed of light.
- B** Thermal radiation is a longitudinal wave.
- C** Thermal radiation travels as an ultra-violet wave.
- D** White surfaces are better emitters of thermal radiation than black surfaces.

**34. M/J/16/12/Q15**

A slice of bread is placed under a red-hot electric grill to make toast.



How does heat energy reach the bread?

- A** conduction and convection
- B** conduction only
- C** convection and radiation
- D** radiation only

**35. O/N/15/12/Q20**

Which statement is correct?

- A** Infra-red radiation cannot travel in a vacuum.
- B** Infra-red radiation cannot travel in solids or in gases.
- C** Infra-red radiation can only travel in a vacuum.
- D** Infra-red radiation can travel in a vacuum and in gases.

**36. M/J/15/11/Q15**

A solid bar is heated at one end.

How is thermal energy transferred to the other end of the bar?

- A** Heated molecules move along the bar, carrying energy to the other end.
- B** Heated molecules move along the bar, giving energy to others along the bar.
- C** Heated molecules stay completely still, but give energy to other molecules.
- D** Heated molecules vibrate more rapidly and pass energy to other molecules.

**37. M/J/15/11/Q16**

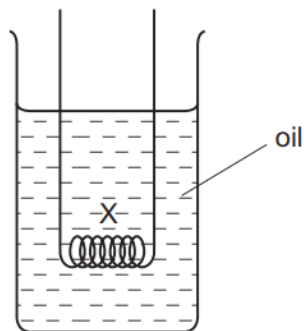
The tubes inside solar heating panels use the Sun's radiation to warm water.

Why are the tubes painted black?

- A** Black surfaces absorb radiation well.
- B** Black surfaces conduct heat well.
- C** Black surfaces emit radiation well.
- D** Black surfaces reflect radiation well.

**38. M/J/15/12/Q15**

An electrical heater is placed in a beaker of cold oil, as shown.



The heater is switched on.

What happens to the liquid at X?

- A** It becomes less dense and falls.
- B** It becomes less dense and rises.
- C** It becomes more dense and falls.
- D** It becomes more dense and rises.

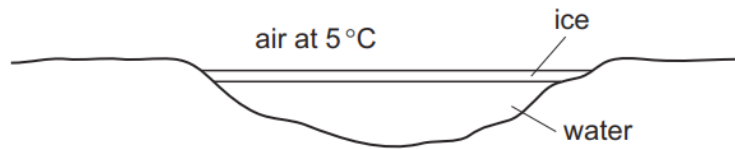
**39. O/N/14/12/Q11**

Which statement about copper explains why it is a better conductor of heat than glass?

- A** Atomic vibration is passed on to neighbouring copper atoms quickly.
- B** Atoms move through the copper and pass on kinetic energy.
- C** There are density changes within the copper.
- D** There are free electrons in the copper.

**40. O/N/14/12/Q12**

The diagram shows a frozen pond with the surface of the ice slowly melting as heat is transferred from the warmer air above it.

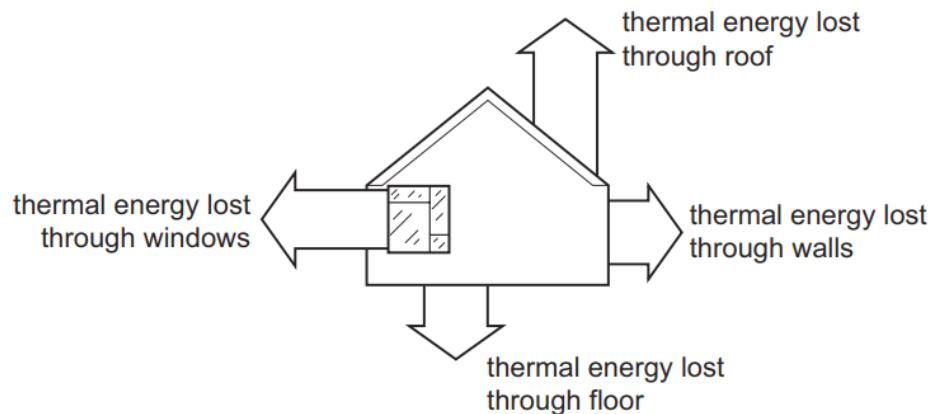


By which processes is heat transferred from the air to the ice?

- A** conduction, convection and radiation
- B** conduction and convection only
- C** convection and radiation only
- D** radiation and conduction only

**41. O/N/13/11/Q13**

On a cold afternoon, a house loses 54 MJ of thermal energy (heat) to its surroundings as shown.



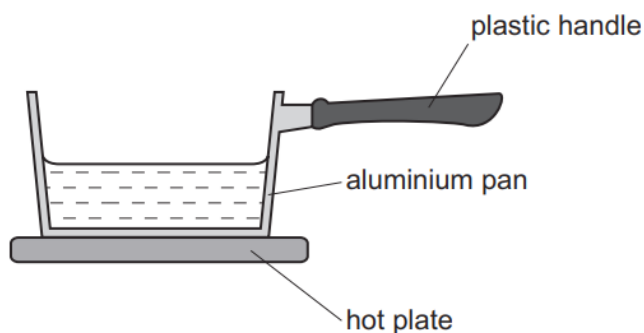
The heating system must supply more than 54 MJ of thermal energy to keep the temperature of the house constant.

Which statement explains this?

- A** The extra thermal energy is lost from the house to the surroundings by other means.
- B** The extra thermal energy keeps the house warmer than the surroundings.
- C** The temperature of the surroundings decreases continuously during this period.
- D** The thermal insulation of the roof is extremely ineffective.

**42. O/N/13/12/Q15**

A saucepan is used to heat up some water.



How is heat transferred through the aluminium pan and through the plastic handle?

|          | heat is transferred through the aluminium pan by          | heat is transferred through the plastic handle by             |
|----------|-----------------------------------------------------------|---------------------------------------------------------------|
| <b>A</b> | the movement of free electrons and the vibration of atoms | the movement of free electrons and the vibration of molecules |
| <b>B</b> | the movement of free electrons and the vibration of atoms | the vibration of molecules only                               |
| <b>C</b> | the movement of free electrons only                       | the movement of free electrons and the vibration of molecules |
| <b>D</b> | the movement of free electrons only                       | the vibration of molecules only                               |

**43. O/N/13/12/Q16**

A silver cup is filled with boiling water from a kettle.

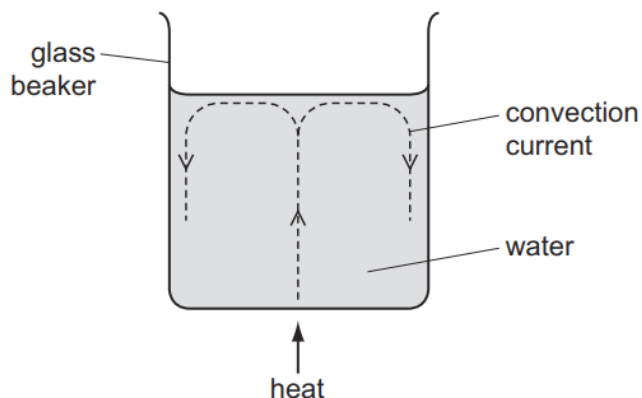
A man touches the outside surface of the cup and finds that it is extremely hot.

Why is the surface so hot?

- A** Convection takes place in the boiling water.
- B** Silver is a good conductor of heat.
- C** The boiling water gives out latent heat.
- D** The shiny surface is a good emitter of infra-red radiation.

**44. M/J/13/11/Q13**

A glass beaker contains water. When the centre of the base of the beaker is heated, a convection current is set up.

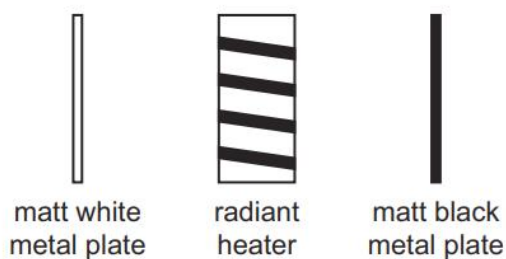


Which statement explains this?

- A** The evaporation of water causes water molecules to rise to the surface.
- B** The expansion of water molecules causes them to rise to the surface.
- C** The water above the heat source rises because it becomes less dense.
- D** The water at the sides sinks because it becomes less dense.

**45. M/J/13/11/Q14**

Two identical metal plates are painted, one matt (dull) white and the other matt black. These are placed at equal distances from a radiant heater as shown. The heater is turned on for five minutes.

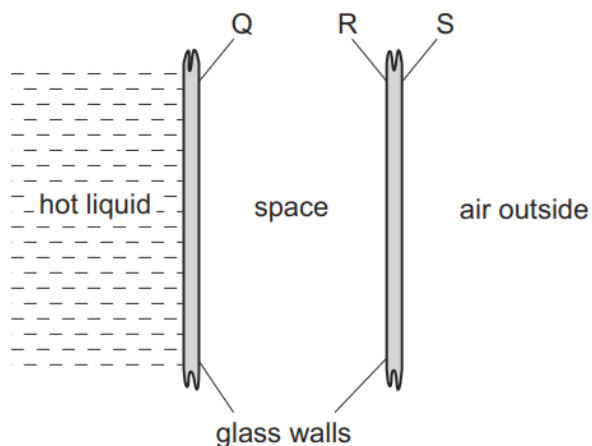


Which metal plate absorbs more energy and which plate emits more energy in this time?

|          | absorbs more | emits more |
|----------|--------------|------------|
| <b>A</b> | black        | black      |
| <b>B</b> | black        | white      |
| <b>C</b> | white        | black      |
| <b>D</b> | white        | white      |

**46. O/N/12/11/Q17**

A student uses a double-walled glass vessel to contain a hot liquid.



What reduces the heat loss by radiation?

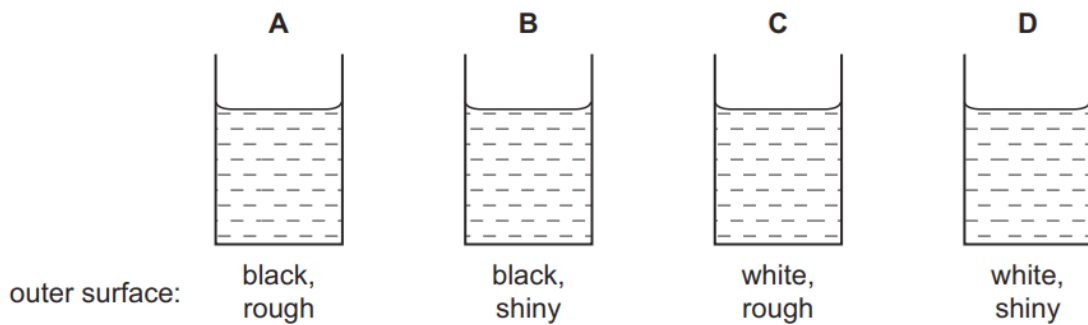
- A** a vacuum in the space between the walls
- B** painting surface Q black
- C** painting surface R black
- D** painting surface S silver

**47. O/N/12/12/Q17**

Four metal cans are identical except for the colour and the texture of their outer surfaces.

100 cm<sup>3</sup> of water at 70 °C is poured into each can.

Which cools the most rapidly?



**48. M/J/12/11/Q17**

A rod of metal is heated at one end.

Which statement best describes the conduction of heat through the metal?

- A** Atoms move from the hot end and hit electrons at the cold end.
- B** Atoms vibrate and hit atoms at the cold end.
- C** Free electrons move from the hot end and hit atoms further along the rod.
- D** Free electrons vibrate and pass energy to free electrons further along the rod.

**49. M/J/12/12/Q17**

How is heat conducted in a metal?

- A** by movement of electrons through the metal only
- B** by movement of atoms through the metal only
- C** by vibration of atoms and movement of electrons through the metal
- D** by vibration of atoms only

**50. O/N/11/11/Q15**

How does heat transfer through a vacuum take place?

- A** by conduction, convection and radiation
- B** by conduction only
- C** by convection only
- D** by radiation only

**51. M/J/11/11/Q16**

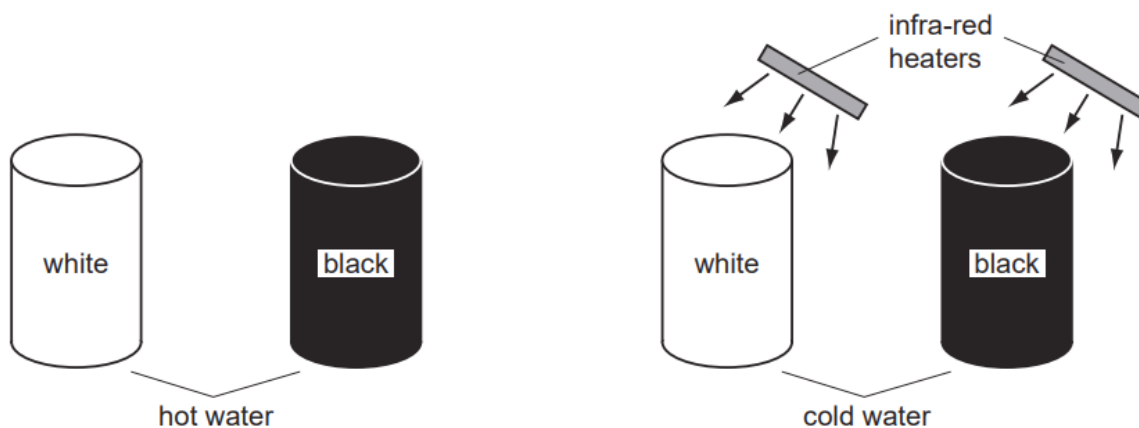
A balloon filled with air is gently heated.

What happens to the mass and the density of the air inside the balloon?

|          | mass           | density        |
|----------|----------------|----------------|
| <b>A</b> | decreases      | decreases      |
| <b>B</b> | decreases      | stays the same |
| <b>C</b> | stays the same | decreases      |
| <b>D</b> | stays the same | stays the same |

**52. O/N/09/Q14**

The diagrams show four cans in a cool room. They are painted as shown. One pair is filled with hot water and left to cool down. The other pair is filled with cold water and placed near infra-red heaters.



The hot water in the black can cools more quickly than the hot water in the white can. The cold water in the black can heats up more quickly than the cold water in the white can.

Which row shows the reasons for this?

|          | better emitter of infra-red | better absorber of infra-red |
|----------|-----------------------------|------------------------------|
| <b>A</b> | black                       | black                        |
| <b>B</b> | black                       | white                        |
| <b>C</b> | white                       | black                        |
| <b>D</b> | white                       | white                        |

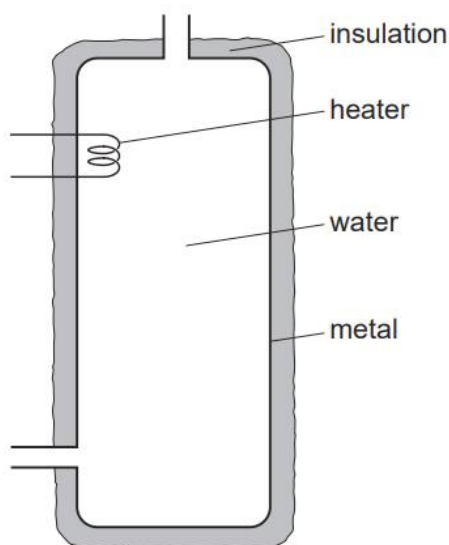
**53. M/J/08/Q14**

In a vacuum flask, which methods of heat transfer are prevented by the vacuum?

- A** conduction only
- B** convection only
- C** conduction and convection only
- D** conduction, convection, and radiation

**54. O/N/07/Q18**

Water at the top of a hot-water tank is heated and the water becomes hot. No water enters or leaves the tank.



Water at the bottom of the tank stays cold for some time.

Why is this?

- A** Cold water at the top of the tank falls to the bottom.
- B** Hot water at the bottom of the tank rises to the top.
- C** Water is a poor conductor of heat.
- D** The insulation is a poor conductor of heat.

**55. O/N/06/Q18**

The heat from the hot water in a metal radiator passes through the metal and then spreads around the room.

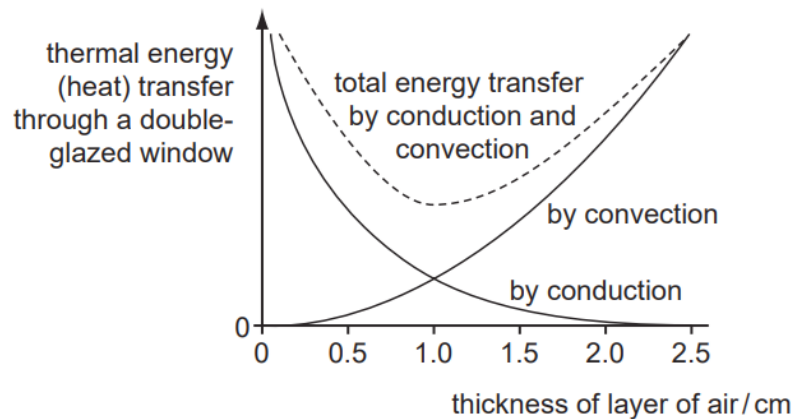
What are the main processes by which the heat is transferred through the radiator and then spread around the room?

|          | through the metal radiator | around the room |
|----------|----------------------------|-----------------|
| <b>A</b> | conduction                 | conduction      |
| <b>B</b> | conduction                 | convection      |
| <b>C</b> | radiation                  | conduction      |
| <b>D</b> | radiation                  | convection      |

**56. M/J/07/Q18**

A double-glazed window has two sheets of glass separated by a layer of air.

Thermal energy is conducted and convected through the layer of air. The amount of conduction and convection varies with the thickness of the layer of air, as shown in the graph.



Which thickness of air produces the smallest energy transfer, and why?

- A** 0.5 cm because there is little convection
- B** 1.0 cm because the total thermal energy transfer is least
- C** 1.5 cm because the total thermal energy transfer is small and conduction is low
- D** 2.0 cm because there is little conduction

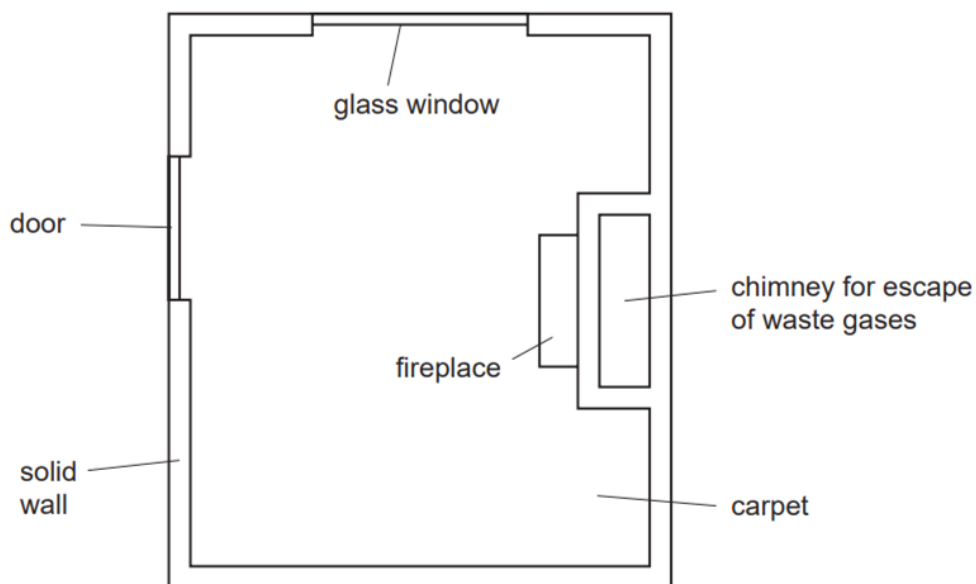
**57. M/J/06/Q20**

Density changes are responsible for which method of thermal energy transfer?

- A** conduction only
- B** convection only
- C** radiation only
- D** conduction, convection and radiation

**58. O/N/05/Q18**

The diagram shows a room seen from above. It is cold outside the room. The room is heated by a small fire in the fireplace.

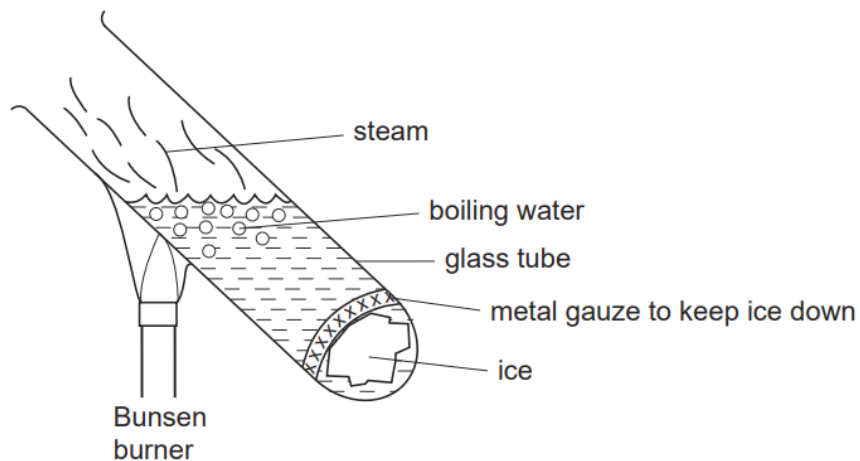


Where is most heat lost by convection?

- A** carpet
- B** chimney
- C** glass window
- D** solid wall

**59. O/N/05/Q19**

An experiment is carried out as shown in the diagram.



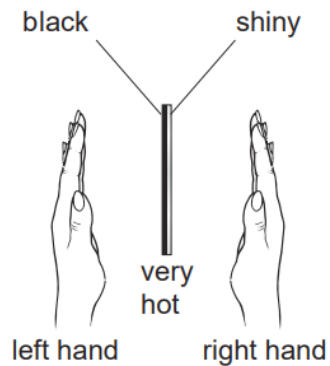
Why does the ice take a long time to melt, even though the water at the top of the tube is boiling?

- A** Convection never occurs in water.
- B** Ice is a poor conductor of heat.
- C** The gauze prevents the energy reaching the ice.
- D** Water is a poor conductor of heat.

**60. M/J/05/Q18**

The diagram shows a thick copper plate that is very hot. One side is black, the other is shiny.

A student places her hands the same distance from each side as shown.



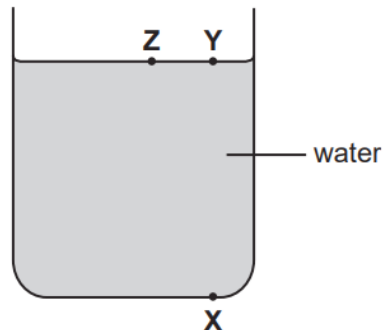
Her left hand feels warmer than her right hand.

Which statement is the correct conclusion from the experiment?

- A** The black side is hotter than the shiny one.
- B** The black side radiates more heat.
- C** The shiny side radiates more heat.
- D** The shiny side is cooling down faster than the black side.

**61. O/N/04/Q21**

A teacher has a large tank of water in which he wants to set up a convection current.

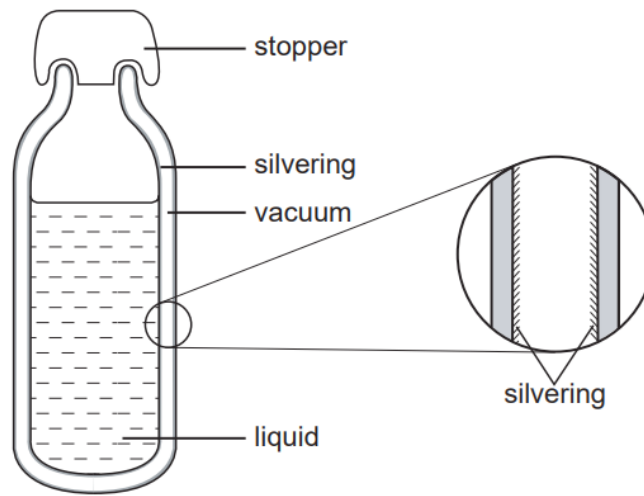


Which of the following arrangements would do this?

- A** cooling at **X**
- B** cooling at **Y**
- C** heating at **Y**
- D** heating at **Z**

**62. M/J/04/Q18**

The diagram shows a vacuum flask and an enlarged view of a section through the flask wall.



The main reason for the silvering is to reduce heat transfer by

- A** conduction only.
- B** radiation only.
- C** conduction and convection.
- D** convection and radiation.